

Context-Aware Selection of AI-generated Audio Descriptions for Blind and Low Vision Users (Final Report)

Melanie Medina

Data Science Major

melaniemedina2028@u.northwestern.edu

Faculty mentor: Shreeya Behera

Department of Statistics and Data Science, Assistant Professor of Instruction

shreeya.behera@northwestern.edu

Project Description

This project began as an exploration into an issue that bothered me. I noticed that video and photo description tools existed as accessibility aids for blind and low vision people, but they relied on APIs, had limited capabilities, and could not handle TV shows. I looked into audio description and realized that Netflix provides human AD, but only for the most popular shows. I was curious whether neural networks could do for AD what auto-captions had done for subtitles. When I looked into AI-generated AD, I found the field was quite behind. There were few publications, and the ones that existed were niche research projects that were usually not open source.

I found one pipeline, Shot-by-Shot, that seemed promising (Xie et al., 2025). It used open-source models I could run locally and was designed for TV show data with ground truth human AD annotations. I used this pipeline on Quest HPC to generate AD for 2,983 short scenes from live-action TV shows, producing five candidate descriptions per scene. I then extracted textual features from those candidates and trained a lightweight selector model to pick the best description among the five, achieving a 13.6 percentage point improvement over a baseline of random selection. I found that action verbs were one of the most important textual features.

The repository for this project, along with generated AD video examples, can be found here: https://github.com/mmelliem/AD_selection.

Project Progress

The first major change was the generation pipeline. I originally intended to use VideoA11y, but found it was built on GPT-4V and that would have cost a significant amount in API calls at scale (Li et al., 2025). It was also designed for YouTube content which was not my focus. Shot-by-Shot used open-source models and was designed specifically for movies and TV shows, so it was a better fit. I also changed my dataset to TV-AD, which is derived from TVQA and compatible with Shot-by-Shot (Lei et al., 2018).

The next significant change was that Phase A took much longer than expected. Generating text descriptions from video was far more computationally expensive than I initially anticipated, so I got access to Quest with my mentor's help. Shot-by-Shot is built on two large models that required powerful GPUs: Qwen2-VL-7B, a vision-language model (Wang et al., 2024), and LLaMA 3 8B, a language model (AI@Meta, 2024). The advantage of using these models was that they were comparable in capability to GPT-4V while being completely open-source and runnable locally.

Another change was in how I generated multiple candidate descriptions. I had originally proposed generating three different descriptions using different prompts, but Shot-by-Shot uses a single hard-coded prompt, making that approach infeasible. So instead, I generated five candidate descriptions per scene by adjusting commands the authors had provided for the LLaMA stage. Even though the descriptions came from the same model, they were noticeably different.

I was also not able to extract visual and audio features as outlined in my proposal. I ran into several incompatibility issues with my dataset and could not align the audio to the video segments. Instead, I used sentence-transformers and spaCy to extract textual features from the generated AD candidates. The features I used from the original proposal were `scene_dur`,

n_tokens, and TTS_speech_dur. I then added the following textual features: n_action_verbs, pronoun_to_noun_ratio, verb_density (action verbs / tokens), noun_ratio (nouns / tokens), type_token_ratio (unique words / total words), len_diff_from_mean (AD length - mean length of all five ADs), consensus_score (cosine similarity between all five AD text embeddings).

For Phase B, I ran the following regression models to predict the value of quality score: Ridge, Lasso, KNN, Random Forest, XGBoost. In addition, I also framed AD selection as a classification task. For each scene, the candidate with the highest quality score was labeled as the target (1) and the rest as 0. I trained logistic regression, random forest, and XGBoost models using 5-fold cross-validation, with hyperparameter tuning via grid search. I evaluated two definitions of quality score, as I found a notable difference between them:

- Quality Score 1 (Q1): Embedding similarity: cosine similarity between candidate AD and human AD sentence embeddings.
- Quality Score 2 (Q2): Action score: the evaluation metric provided by Shot-by-Shot authors, which combines semantic similarity and verb lemma matching between candidate and ground truth, and removes character name bias. This metric was shown by the authors to correlate well with human judgments.

I was not able to complete the second stage of evaluation with the user feedback study due to difficulties with IRB paperwork and participant recruitment. I only completed the statistical evaluation outlined in my proposal.

Project Results

Results are reported as selector accuracy, or the proportion of scenes where the model correctly identifies the candidate with the highest quality score.

Model	Q1 acc.	Q2 acc.
Baseline: random	20.1%	20.1%
Baseline: shortest	21.9%	18.0%
Baseline: first	23.2%	22.2%
Classification		
Logistic Regression	25.0%	33.7% ✓
Random Forest	23.8%	31.0%
XGBoost	25.5%	31.8%
Regression		
Ridge	20.7%	29.3%
Lasso	20.5%	30.0% ✓
KNN	21.3%	24.5%
Random Forest	20.7%	29.7%
XGBoost	19.7%	29.3%

Logistic regression using action score (Q2) as the target achieved 33.7% selector accuracy, an 13.6 percentage point improvement over the baseline of random chance, which is within the proposed 10–15 percentage point threshold from my original proposal. Q1 (embedding similarity) performed closer to chance regardless of model, which pointed to a fundamental issue with that metric. Logistic regression outperforming random forest and XGBoost was unexpected, but likely reflects the small feature set and a lesser degree of overfitting.

Two other notable patterns are that regression-based selection (30.0%) nearly matches classification-based selection (33.7%), and that all regression models converged to similar accuracy under Q2, confirming that the bottleneck is the feature set rather than model architecture.

The consistent failure of Q1 across all models reveals that raw cosine similarity between AI-generated and human AD is not an effective quality metric for this task. Q2 incorporates both semantic similarity and verb coverage, which makes a meaningful difference.

Feature Importances

The most informative features under Q2 were:

n_action_verbs	0.253
pronoun_to_noun_ratio	0.224
consensus_score	0.203
len_diff_from_mean	0.146
type_token_ratio	0.108
verb_density	0.044

The dominance of action verb count and pronoun to noun ratio makes sense. Since action score is based on matching key action words, candidates who use more of these verbs naturally score higher. Consensus score, or the similarity between candidates, also scoring well highlights that AD candidates that are most representative of the average description can be of higher quality.

Addressing the Research Question

My research question asked which selector model and contextual features are most effective for selecting AI-generated audio descriptions. The answer is that the selector model type does not seem to matter much (logistic regression performs similarly to more complex models). Action verbs appear to be the most important feature, while semantic similarity to human AD and candidate length are less important. How quality is defined also makes a large difference since action score produced far more interpretable and useful results than embedding similarity alone. Since I was not able to complete a user study, it remains unknown whether these computational findings align with what blind and low-vision viewers actually prefer. The Shot-by-Shot authors did demonstrate that action score correlates with human judgments, which is a promising sign (Xie et al., 2025).

Next Steps

A human study is the clearest next step. I would like to obtain IRB approval and use snowball sampling as outlined in my proposal to recruit blind and low vision participants.

I would also like to incorporate richer features. This project was limited to textual features due to dataset constraints, but the original proposal called for audio-derived timing features and

visual event counts. These could meaningfully improve selector accuracy.

Candidate diversity is another limitation. All five candidates came from the same pipeline. Ideally, candidates from multiple generation pipelines would be compared, but this was logistically infeasible as I was not able to set up another working pipeline during the research period.

The longer-term motivation behind this project is the absence of any accessible, open-source tool that non-programmers can use to generate AD for long-form video. Free commercial tools like Microsoft Seeing AI exist but are not designed for narrative content. Shot-by-Shot and VideoA11y are impressive research pipelines that address this gap but require powerful GPUs and technical expertise most users do not have. The selector model from this project is lightweight enough to run on a laptop. The bottleneck is the generation step. Future work I want to explore is whether a lighter generation model could produce passable AD on consumer hardware, with this selector filtering the output.

Academic/Artistic Development

Before this project, I had never worked with a computing service like Quest. Becoming comfortable with debugging in a remote environment, running GPU allocations, and job scheduling will be very useful for my future work. I also learned a lot about machine learning in practice: using VLMs and LLMs locally, performing textual feature extraction, and training classification and regression models on the outputs.

One of the biggest academic lessons was the difficulty of experimental design in a specialized research area. I spent a substantial portion of the project testing alternative pipelines, troubleshooting data format incompatibilities, and dealing with dependency conflicts. Working within an existing research pipeline was valuable, but it also taught me to be more skeptical of evaluation metrics. My original quality score (cosine similarity) seemed reasonable at first and produced results that looked like they were working. It was only by comparing it against an alternative metric produced by other researchers that I realized how much it shapes what a model learns and what conclusions can be drawn.

Personal Development

The biggest personal lesson from this project was becoming more comfortable with uncertainty and with the fact that research rarely follows the original plan. I began the summer with a broad vision, but many parts had to change because of logistical constraints. Working through those constraints made me more independent. I had to investigate alternative pipelines, troubleshoot unfamiliar software and computing environments, make decisions about which parts of the original proposal were still feasible, and determine what conclusions my results actually supported.

The project also changed how I think about accessibility technology. I began with the goal of improving AI-generated audio descriptions, but the experience made me more aware of the gap between a research prototype and a tool that an actual person can use. I am very determined to continue this work and make this technology more usable and accessible.

References¹

- AI@Meta. (2024). *Llama 3 Model Card*. GitHub.
https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md
- Lei, J., Yu, L., Bansal, M., & Berg, T. L. (2018). *TVQA: Localized, compositional video question answering*. arXiv.Org. <https://arxiv.org/abs/1809.01696>
- Li, C., Padmanabhuni, S., Cheema, M., Hasti Seifi, & Fazli, P. (2025). *VideoA11y: Method and Dataset for Accessible Video Description* (2502.20480V1). arXiv.
<https://arxiv.org/abs/2502.20480v1>
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., & Lin, J. (2024). *Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution*. arXiv.Org. <https://arxiv.org/abs/2409.12191>
- Xie, J., Han, T., Bain, M., Arsha Nagrani, Khandelwal, E., Gül Varol, Xie, W., & Zisserman, A. (2025). *Shot-by-shot: Film-grammar-aware training-free audio description generation* (2504.01020V2). arXiv. <https://arxiv.org/abs/2504.01020v2>

¹ The study resulting in this report was assisted by a grant from the BPF Undergraduate Research Grant Award, which is administered by Northwestern University's Department of Statistics and Data Science. However, the conclusions, opinions, and other statements in this report are the author's and not necessarily those of the department or the sponsoring institution.